

The Lattice of Subgroups of $(\mathbb{Z}/28\mathbb{Z})^*$

In class on Wednesday, we began to work out the lattice of subgroups of $(\mathbb{Z}/28\mathbb{Z})^*$, but moved on to an introduction to cosets before finishing our calculation. Here we summarize and complete the lattice.

First, from grade school factoring of small numbers, we clearly have

$$(\mathbb{Z}/28\mathbb{Z})^* = \{1, 3, 5, 9, 11, 13, -13, -11, -9, -5, -3, -1\}$$

(where we allow negative numbers to induce some symmetry and streamline our calculations a bit). Next, let us find all the cyclic subgroups of $(\mathbb{Z}/28\mathbb{Z})^*$. Along the way, we save time by observing that $9 = 3^2$ and thus $\langle 9 \rangle$ will simply consist of half of $\langle 3 \rangle$: the portion corresponding to even powers. Our cyclic subgroups are then the following:

$$\langle 1 \rangle = \{1\}$$

$$\langle -1 \rangle = \{1, -1\}$$

$$\langle 3 \rangle = \{1, 3, 9, -1, -3, -9\}$$

$$\langle 5 \rangle = \{1, 5, -3, 13, 9, -11\}$$

$$\langle 9 \rangle = \{1, 9, -3\}$$

$$\langle 11 \rangle = \{1, 11, 9, -13, -3, -5\}$$

$$\langle 13 \rangle = \{1, 13\}$$

$$\langle -13 \rangle = \{1, -13\}$$

...and the remaining possible generators yield subgroups already in our list: $\langle -11 \rangle = \langle 5 \rangle$, $\langle -9 \rangle = \langle 3 \rangle$, $\langle -5 \rangle = \langle 11 \rangle$, and $\langle -3 \rangle = \langle 9 \rangle$. We thus see that (a) there are exactly 8 distinct cyclic subgroups and (b) $(\mathbb{Z}/28\mathbb{Z})^*$ is *not* cyclic (since all the preceding subgroups have cardinality dividing 6).

The most natural thing to try next is to see if one can express all of $(\mathbb{Z}/28\mathbb{Z})^*$ as a *product* of powers of a *collection of generators*. A simple strategy then, to generate the elements missing from our largest subgroups, would be to take one of our order 6 subgroups and multiply it by an *outside* element to get the other half of our entire group! In particular, it easily checked that $13 \cdot \langle 3 \rangle = \{13 \cdot 1, 13 \cdot 3, 13 \cdot 9, 13(-1), 13(-3), 13(-9)\} = \{13, 11, 5, -13, -11, -5\}$ (using our new coset notation) and thus $\langle 3 \rangle \sqcup (13 \cdot \langle 3 \rangle)$, where $\sqcup$ denotes a disjoint union.

So if we define $\langle 3, 13 \rangle = \{3^a 13^b \mid a, b \in \mathbb{Z}\}$ (in $(\mathbb{Z}/28\mathbb{Z})^*$), we see that $\langle 3, 13 \rangle = (\mathbb{Z}/28\mathbb{Z})^*$. The terminology is that $(\mathbb{Z}/28\mathbb{Z})^*$ is *generated by* $\{3, 13\}$. Now, knowing that it is possible to have groups that require *more* than one element to generate them, let us search for more subgroups: Note that $\langle 9, 13 \rangle = \langle 9 \rangle \sqcup (13 \cdot \langle 9 \rangle) = \{1, 9, -3, 13, 5, -11\} = \langle 5 \rangle$. Also, $\langle 1, 13 \rangle = \langle 13 \rangle$, $\langle 13, 13 \rangle = \langle 13 \rangle$, $\langle 5, 13 \rangle = \langle 5 \rangle$ (since $13 \in \langle 5 \rangle$), and $\langle 11, 13 \rangle = (\mathbb{Z}/28\mathbb{Z})^*$ (since $\langle 11 \rangle$ has order 6 and $13 \notin \langle 11 \rangle$). So we see no new subgroups yet.

However, one can easily check that $\langle -1, 13 \rangle = \langle -13, 13 \rangle = \{1, 13, -1, -13\}$, which has cardinality 4. In particular, it easily checked that $\langle -1, 13 \rangle$ is isomorphic to the Klein 4-group

That we at last have *all* possible subgroups is easily checked if we know *Lagrange's Theorem* (which we will cover on Thursday, Sep. 20, 2018): Any subgroup of $(\mathbb{Z}/28\mathbb{Z})^*$ must have order dividing 12 (the order of $(\mathbb{Z}/28\mathbb{Z})^*$). So we merely need to check if adding more generators to any of our 10 subgroups so far yields any new subgroups. We leave this as an exercise for the reader, and thus arrive at our lattice of subgroups:

